

Share Your Knowledge

If you are an expert in your field of electronics, why not share your knowledge with the world?



Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

Some of the topics we want you to write about are:

- Photonics: Optics & Optoelectronics
- Automation
- Power Electronics
- Components
- Telecom

So if you have an interesting topic or original article to share, please let us know at efyeditorial@efy.in

We would love to hear your good ideas too!

electronics
FOR YOU

Its many forms...

In his article titled 'Electronics for the Human Body,' Dr John A. Rogers of University of Illinois, Urbana-Champaign, explains how recent advancements have managed to bridge the gap between the soft, curvilinear and continuously-evolving human body and devices that are rigid, planar and physically static. He goes on to categorise modern medical electronics into three approaches:

Soft electronics. This involves configuring hard, inorganic functional materials into thin, open-mesh micro-architectures and embedding these in soft, elastomeric films, so that the resulting electronics can be bent, twisted or pulled without affecting their functionality or performance. When implemented with bio-compatible interface materials, the electronics can be integrated with organ systems such as the brain, heart and skin.

Injectable electronics. One step ahead, this involves delivering electronics further into the depth of the body, and not just on the surface. Flexible filaments serve as support for micro-scale device components that can be transported to targeted regions in a minimally-invasive fashion using thin, releasable injection needles.

Bio-resorbable electronics. Releasing injection needles in the previous case involves bio-resorption of a thin adhesive layer upon contact with cerebro-spinal fluid. Recently, it has been found that this concept of bio-resorption can be extended to entire functional systems, where all active and passive materials dissolve completely in a controlled fashion at programmable rates when immersed in bio-fluids. This opens the possibility of advanced devices for drug release, which provide high performance and stable operation for a specified time period and then completely dissolve, rather than stay on in the body.

—Adapted from: rogers.matse.illinois.edu/files/2015/jamabioe.pdf

A series of small devices being developed by CellScope, a spin-off from UC Berkeley, is also expected to bring advanced diagnostics home to common folks. Oto Home and Oto Clinic, for example, are small devices that fit onto the camera of a smartphone, enabling users to capture images of the ear's membranes, to detect ear infections. The idea behind mobile phone microscopy, a concept pioneered by Fletcher Lab of UC Berkeley, is to develop small devices that add optics, illumination and hardware automation to the cameras of regular mobile phones and tablets, to create mobile microscopes and diagnostic solutions.

More on the anvil

The future holds many more such devices, as was seen at Elsevier's 4th International Conference on Bio-Sensing Technology. One of the interesting technologies demonstrated at the event was a home diagnosis card developed in Sweden by Linköping University and Acreo. The device, the size of a credit card, takes a drop of blood or saliva as input from the user and gives diagnostic information on his or her phone. This information can help monitor diabetes, heart and

kidney diseases and even cancer. Screen printing all electronics on the card makes it very cost-effective, too. It currently costs €5 each, but as volume picks up, it can be manufactured at €0.50 per piece.

According to Elsevier's press release on the event, "This means these have the potential to provide patients and doctors in developing countries with accessible, affordable medical tests. For example, the printed card could be made part of the packaging of antibiotics, helping determine which antibiotic would be best to treat a patient's infection. Such printable devices could also be worn like plasters or contact lenses, transmitting information to mobile phones."

Another recent development in this space is a type of e-skin developed by Suzhou Institute of Nano-Tech and Nano-Bionics, Chinese Academy of Sciences, China. Born of nanotechnology and flexible electronics, e-skin has the ability to detect changes in pressure, which can, in turn, be used to monitor blood pressure, heart rate and wrist pulse on real-time basis. The team has worked to make the sensor element very sensitive, while making the material very flexible. This they have done